

Abstract

Web Scraping and Product Price Analysis

E-commerce platforms provide a large volume of product information through online listings, but comparing air conditioner prices, brands, discounts, and specifications from scattered product pages is difficult through manual analysis. This project analyzes 720 Amazon India Air Conditioner listings collected through web scraping, containing product details such as name, ratings, number of ratings, selling price, actual price, tonnage, energy star rating, discount percentage, and brand information. Using Python, Pandas, NumPy, and Matplotlib, the dataset is cleaned and preprocessed by converting text-based prices and ratings into numerical values, auditing missing and placeholder values, standardising brands, extracting tonnage and energy-star information, and removing 68 non-AC items such as stands, motors, and portable coolers. Exploratory Data Analysis (EDA) using descriptive statistics, grouped comparisons, and visualizations is performed to study price distributions, brand-wise pricing, tonnage-wise trends, energy-star shares, discount patterns, and the relationship between price and ratings. The final analysis considers 652 genuine AC listings, including 423 with verified prices and 392 with verified ratings, providing clear insights into product pricing and market patterns. The project aims to support buyers and sellers in understanding AC price differences, discounts, brand trends, and product characteristics, while demonstrating an effective Python-based workflow for web-scraped e-commerce data analysis.

Team no: 04

25B11CS608 – M. Akshaya

25B11CS710 – P. Jagannadha Veera Manikanta

25B11CS388 – K. Sai Krishna

25B11CS923 – T. Yeghna Surya Teja